3B 1 COVALENT BONDING

SPECIFICATION REFERENCE 3.10

LEARNING OBJECTIVES

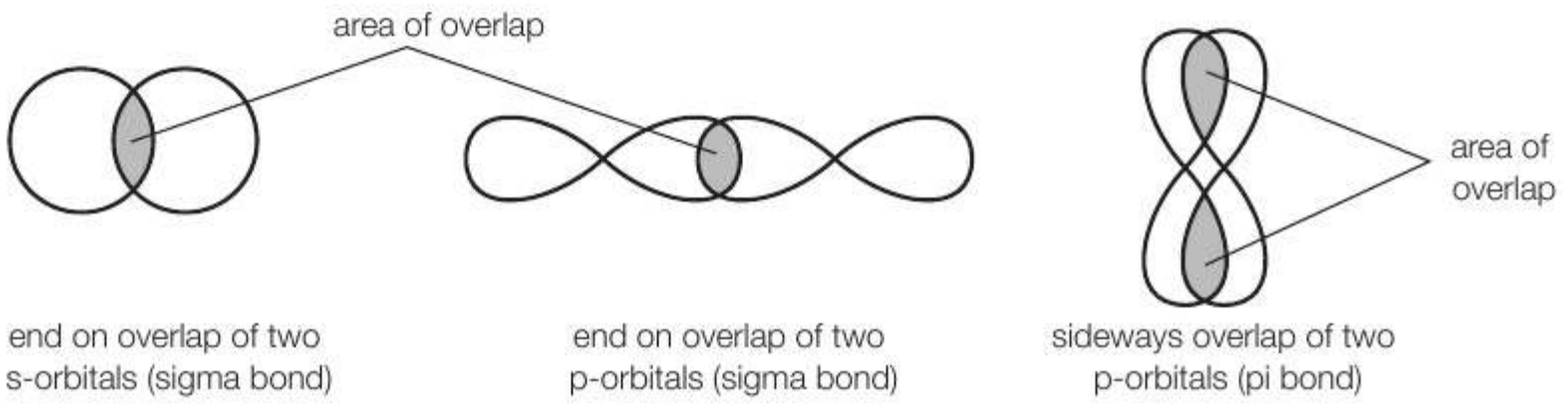
- Know that a covalent bond is formed by the overlap of two atomic orbitals each containing a single electron.
- Know that a covalent bond is the strong electrostatic attraction between the nuclei of two atoms and the bonding (shared) pair of electrons.
- Understand the relationship between bond length and bond strength for covalent bonds.

FORMATION OF COVALENT BONDS

A covalent bond forms between two atoms when an atomic orbital containing a single electron from one atom overlaps with an atomic orbital, which also contains a single electron, of another atom. The two electrons in the area of overlap are the bonding electrons. They are sometimes referred to as a 'shared pair of electrons'. The covalent bond is the electrostatic attraction between the two nuclei of the bonded atoms and the pair of electrons shared between them.

The atomic orbitals involved can be any of those found in the atoms, but we shall limit our discussion to those involving only s- and p-orbitals.

Fig A shows three ways in which these orbitals may overlap.



▲ **fig A** Formation of sigma bonds by end-on overlap of atomic orbitals and a pi bond by sideways overlap of p-orbitals.

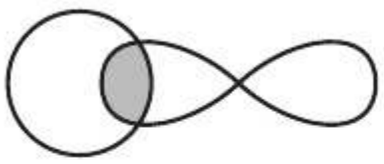
An end-on overlap leads to the formation of sigma (σ) bonds. This leads to the formation of a single covalent bond between the two atoms.

A sideways overlap of two p-orbitals leads to the formation of a pi (π) bond. A feature of a π bond is that it cannot form until a σ bond has been formed. For this reason π bonds only exist between atoms that are joined by double or triple bonds.

The different types of orbital overlap are shown in the following examples.

DID YOU KNOW?

There is a fourth way that overlap can occur. An s- and p-orbital can overlap end on.



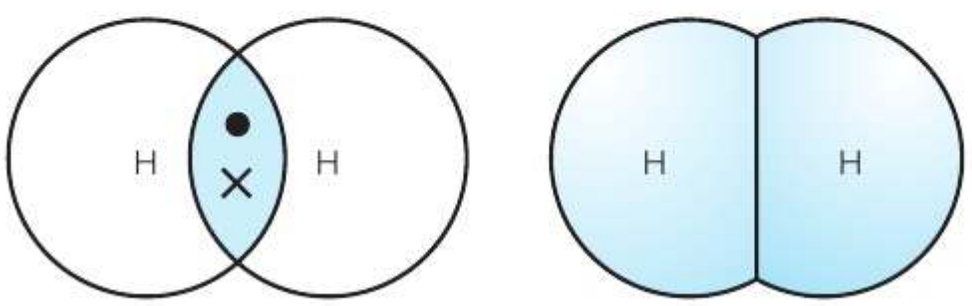
▲ **fig B** Overlap of an s- and a p-orbital to form a sigma bond.

This overlap, however, can only result from atoms of two different elements. This almost always leads to the formation of a variant of a covalent bond, known as a 'polar' covalent bond (see Topic 3B.2).

EXAMPLE 1. HYDROGEN

A hydrogen atom has an electronic configuration of $1s^1$.

When two hydrogen atoms bond together to form a hydrogen molecule, the two s-orbitals overlap to form a new molecular orbital. The two electrons then exist in this new orbital. The highest electron density is between the two nuclei.



▲ **fig C** Formation of the σ bond in hydrogen.

EXAMPLE 2. CHLORINE

A chlorine atom has an electronic configuration of $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p_x^2\ 3p_y^2\ 3p_z^1$.

When two chlorine atoms bond together, the two p orbitals (each containing a single electron) overlap.

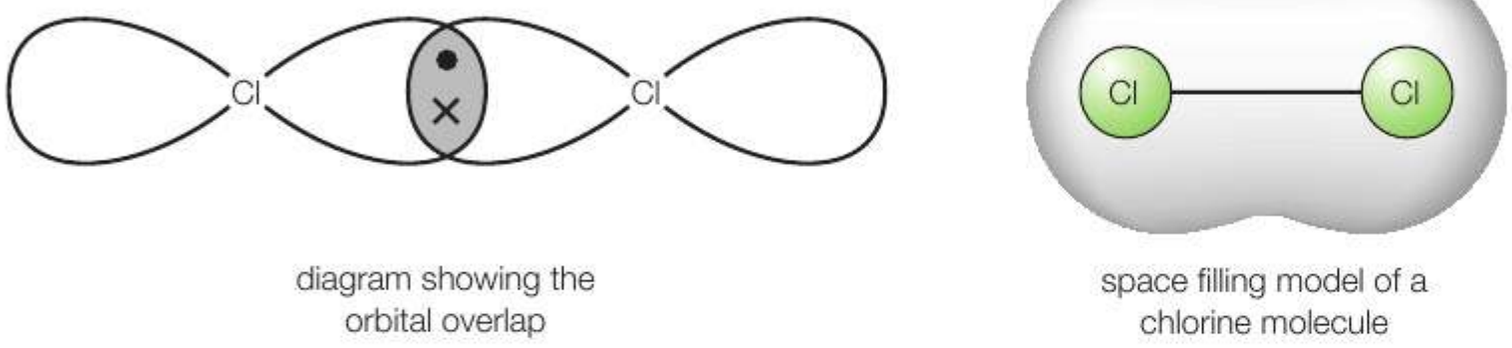


diagram showing the orbital overlap

space filling model of a chlorine molecule

▲ **fig D** Formation of the σ bond in chlorine.

DID YOU KNOW?

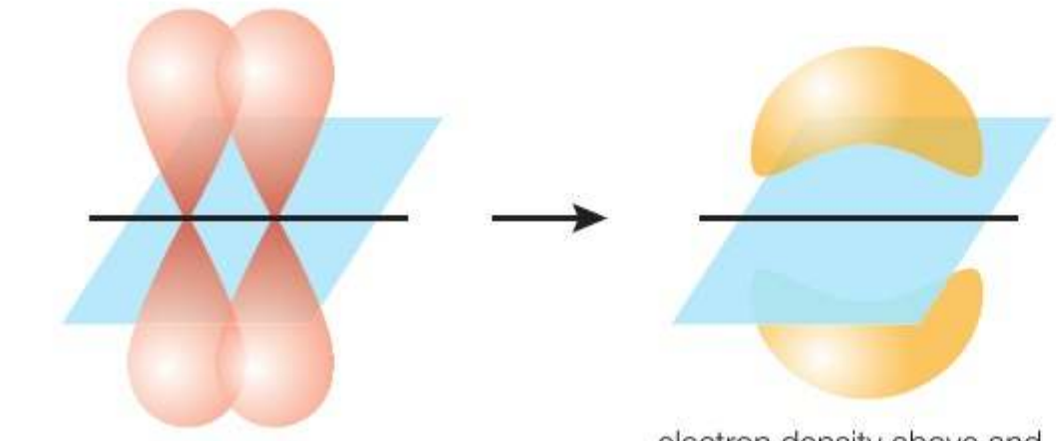
This view of the bonding in chlorine is very simple.

An alternative theory describes the bonding as the overlap between two sp^3 hybrid orbitals. This theory is not included in this book, but we would encourage you to carry out your own research if you are interested. Look up 'orbital hybridisation'.

EXAMPLE 3. π BOND FORMATION

Once a σ bond has been formed, it is possible, in certain circumstances, for a π bond to form.

The π bond results in a high electron density both above and below the molecule, as shown in **fig E**.



sideways overlap of p-orbitals

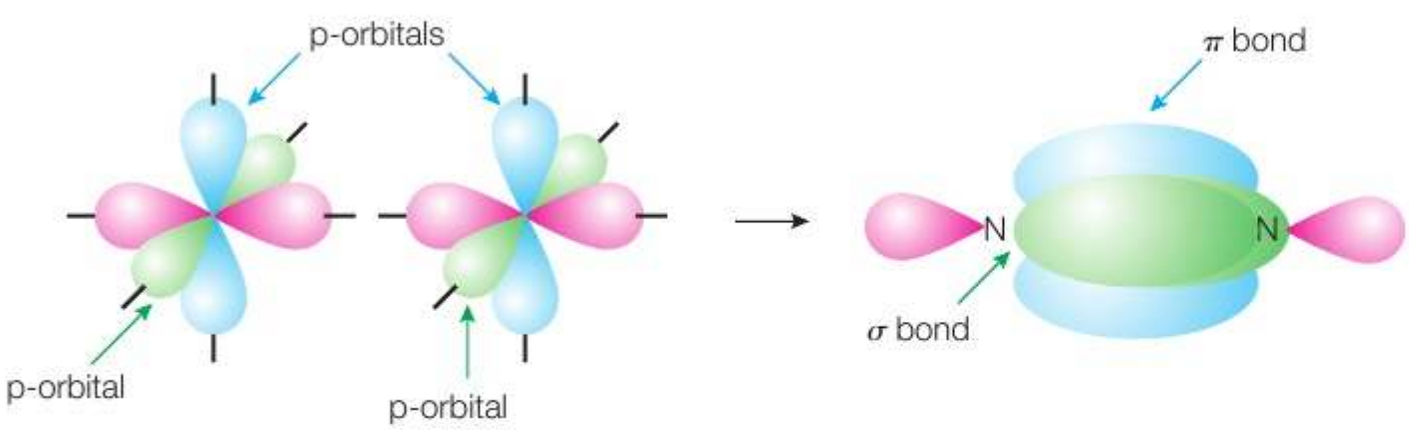
electron density above and below the molecule

▲ **fig E** Formation of a π bond.

This is what happens in the ethene molecule. One of the bonds between the carbon atoms is a σ bond; the other is a π bond.

The π bond in ethene is weaker than the σ bond. This is the reason for the increased reactivity of alkenes compared with alkanes, and why alkenes can easily undergo addition reactions. (See **Topic 5** for more information.)

The triple bond in the nitrogen molecule (N≡N) is made up of one σ bond and two π bonds.



The p-orbitals marked are those that are involved in the formation of the pi bonds.

▲ **fig F** Formation of the two π bonds in nitrogen.

BOND LENGTH AND BOND STRENGTH

The **bond length** is the distance between the nuclei of two atoms that are covalently bonded together.

The strength of a covalent bond is measured in terms of the amount of energy required to break one mole of the bond in the gaseous state (see **Topic 6**).

Table A shows the relationship between the bond length and bond strength of a selection of covalent bonds.

BOND	BOND LENGTH / nm	BOND STRENGTH / kJ mol^{-1}
Cl–Cl	0.199	242
Br–Br	0.228	193
I–I	0.267	151
C–C	0.154	347
C=C	0.134	612
C≡C	0.120	838
N–N	0.145	158
N=N	0.120	410
N≡N	0.110	945
O–O	0.148	144
O=O	0.121	498

table A Relationship between bond length and bond strength for a range of covalent bonds.

The general relationship between bond length and bond strength, for bonds that are of a similar nature, is the shorter the bond, the greater the bond strength. This is a result of an increase in electrostatic attraction between the two nuclei and the electrons in the overlapping atomic orbitals.

LEARNING TIP

When making a comparison between bond length and bond strength, it is important to compare 'like with like', in other words, compare things that are the same or very similar.

For example, the strength of the C–C bond (347 kJ mol^{-1}) is greater than that of the N–N bond (158 kJ mol^{-1}) despite being longer (0.154 nm compared with 0.145 nm).
In a molecule such as hydrazine ($H_2N–NH_2$), each nitrogen atom has a non-bonding (lone) pair of electrons and these repel one another, weakening the bond.
In a molecule such as ethane ($H_3C–CH_3$), the carbon atoms do not have any lone pairs.

CHECKPOINT

1. Suggest a reason for the following trend in bond strengths:

   C–C > Si–Si > Ge–Ge

2. The F–F bond in fluorine is much shorter (0.142 nm) than the Cl–Cl (0.199 nm) bond in chlorine, and yet it is much weaker (158 kJ mol^{-1} compared with 242 kJ mol^{-1}). Suggest a reason for this.
3. Suggest a reason why the sigma (σ) bond between the two carbon atoms in the ethene molecule is stronger that the pi (π) bond.

SUBJECT VOCABULARY

bond length the distance between the nuclei of two atoms that are covalently bonded together

3B 2 ELECTRONEGATIVITY AND BOND POLARITY

SPECIFICATION REFERENCE: 3.10(ii) 3.13 3.14

LEARNING OBJECTIVES

- Know that electronegativity is the ability of an atom to attract a bonding pair of electrons.
- Know that ionic and covalent bonding are the extremes of a continuum of bonding type and that electronegativity differences lead to bond polarity.
- Understand what a polar covalent bond is.
- Understand that electron density maps for discrete (simple) molecules show that there is a high electron density between the nuclei of two covalently bonded atoms.

WHAT IS ELECTRONEGATIVITY?

Electronegativity is the ability of an atom to attract a bonding pair of electrons.

The electronegativity of elements, in general:

- decreases down a group of the Periodic Table, that is, from top to bottom
- increases from left to right across a period.

This is demonstrated in the following section of the Periodic Table (**fig A**).

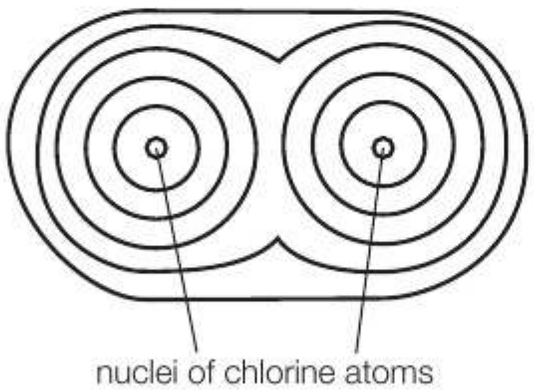
								H 2.1									He
Li 1.0	Be 1.5											B 2.0	C 2.5	N 3.0	O 3.5	F 4.0	Ne
Na 0.9	Mg 1.2											Al 1.5	Si 1.8	P 2.1	S 2.5	Cl 3.0	Ar
K 0.8	Ca 1.0	Sc 1.3	Ti 1.5	V 1.6	Cr 1.6	Mn 1.5	Fe 1.8	Co 1.8	Ni 1.8	Cu 1.9	Zn 1.6	Ga 1.6	Ge 1.8	As 2.0	Se 2.4	Br 2.8	Kr
Rb 0.8	Sr 1.0	Y 1.2	Zr 1.4	Nb 1.6	Mo 1.8	Tc 1.9	Ru 2.2	Rh 2.2	Pd 2.2	Ag 1.9	Cd 1.7	In 1.7	Sn 1.8	Sb 1.9	Te 2.1	I 2.5	Xe
Cs 0.7	Ba 0.9	La 1.1	Hf 1.3	Ta 1.5	W 1.7	Re 1.9	Os 2.2	Ir 2.2	Pt 2.2	Au 2.4	Hg 1.9	Tl 1.8	Pb 1.8	Bi 1.9	Po 2.0	At 2.2	Rn

▲ **fig A** able of electronegativities.

DISTRIBUTION OF ELECTRON DENSITY

If two atoms of the same element are bonded together by the overlap of atomic orbitals, the distribution of electron density between the two nuclei will be symmetrical. This is because the ability of each atom to attract the bonding pair of electrons is identical.

The diagram in **fig B** is an electron density map for chlorine (Cl_2):

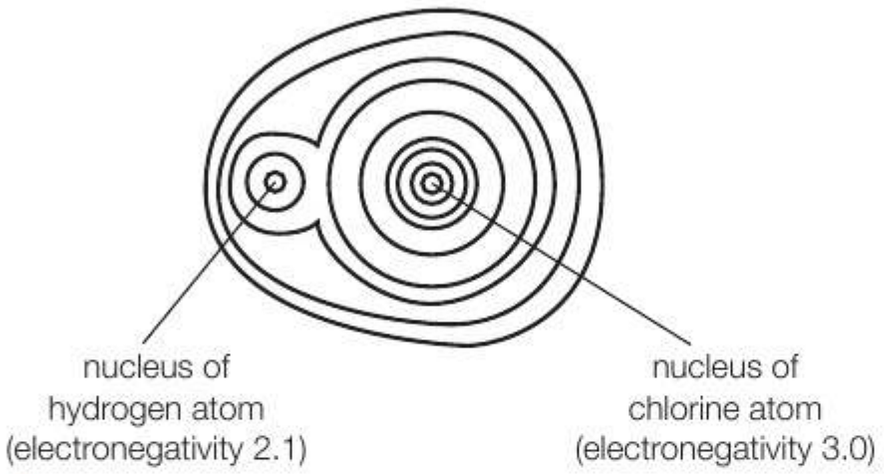


▲ **fig B** Electron density map of a chlorine molecule.

The diagram looks like a contour map. The contour lines correspond to electron density. You can think of them as showing how likely it is that a bonding electron will fall within that contour at a given instant in time. For a normal covalent bond, the contours are symmetrical around the nuclei.

POLAR COVALENT BONDS

However, if the two atoms bonded together are from elements that have different electronegativities, then the distribution of electron density will not be symmetrical about the two nuclei. This is shown in **fig C** by the electron density map for the hydrogen chloride (HCl) molecule.



▲ **fig C** Electron density map of a hydrogen chloride molecule.

The contour lines are more closely spaced near to the chlorine atom, which is the atom with the higher electronegativity.

Since the electron density is higher around the chlorine atom, that end of the molecule has acquired a slightly negative charge. This is represented by the symbol δ–. The other end of the molecule carries a slightly positive charge, represented by the symbol δ+.

$$H^{\delta+} — Cl^{\delta-}$$

A bond like this is called a **polar covalent bond** or sometimes just a 'polar bond'.

Another way of representing a polar covalent bond is to use an arrow to show the direction of electron drift.

$$H \rightarrow Cl$$

Other examples of polar covalent bonds are:

$$C^{\delta+} \rightarrow Cl^{\delta-} \qquad H^{\delta+} \rightarrow O^{\delta-} \qquad H^{\delta+} \rightarrow N^{\delta-} \qquad H^{\delta+} \rightarrow C^{\delta-}$$

LEARNING TIP

Covalent bonds that are non-polar are sometimes called 'normal' covalent bonds or 'pure' covalent bonds. This is to distinguish them from polar covalent bonds.

Both types of bond are formed by the overlap of atomic orbitals.

In both cases the overlapping area contains two electrons per bond formed.

CONTINUUM OF BONDING TYPE

Polar covalent bonds can be thought of as being between two ideals of bonding types. These ideals are:

- pure (100%) covalent
- pure (100%) ionic.

Consider a polar covalent bond as a covalent bond that has some degree of ionic character.

If the electronegativity difference is large enough, then the main type of bonding is ionic.

A very approximate measure of the degree of ionic bonding in a compound is given in **table A**.

ELECTRONEGATIVITY DIFFERENCE	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0	1.1
APPROXIMATE % IONIC CHARACTER	0	0.5	1	2	4	6	9	12	15	19	22	26

ELECTRONEGATIVITY DIFFERENCE	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2.0	2.1	2.2	2.3
APPROXIMATE % IONIC CHARACTER	30	34	39	43	47	51	55	59	63	67	70	74

ELECTRONEGATIVITY DIFFERENCE	2.4	2.5	2.6	2.7	2.8	2.9	3.0	3.1	3.2	3.3
APPROXIMATE % IONIC CHARACTER	76	79	82	84	86	88	89	90	91	92

table A Relationship between percentage ionic character and difference in electronegativity.

CHECKPOINT

1. Suggest why the electronegativity of fluorine is greater than that of chlorine, despite the fact that the nucleus of a chlorine atom contains more protons.

2. Ionic bonding and covalent bonding are two extremes of chemical bonding. Many compounds have bonding that is intermediate in character.

(a) Giving an example in each case, explain what is meant by the terms:

(i) ionic bonding, and (ii) covalent bonding.

(b) Select a compound that has bonding of an intermediate character and explain why it has this type of bonding.

3. Place the following bonds in order of decreasing polarity (i.e. place the most polar first).

C–Br C–Cl C–F C–I

Explain how you arrived at your answer.

SUBJECT VOCABULARY

electronegativity the ability of an atom to attract a bonding pair of electrons in a covalent bond

polar covalent bond a type of covalent bond between two atoms where the bonding electrons are unequally distributed; because of this, one atom carries a slight negative charge and the other a slight positive charge

3B 3 BONDING IN DISCRETE (SIMPLE) MOLECULES

3.11(i)

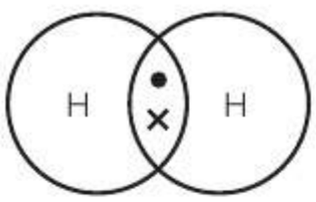
LEARNING OBJECTIVES

- Understand what is meant by the term discrete (simple) molecule.
- Draw dot-and-cross diagrams to show electrons in discrete molecules with single, double and triple bonds.
- Draw displayed formulae to represent the bonding in discrete molecules.

DID YOU KNOW?

Molecules are common in organic substances (and therefore biochemistry). They also make up most of the oceans and the atmosphere.

However, ionic crystals (salts) and giant covalent crystals (network solids) are often made up of repeating unit cells that extend either in a plane (such as in graphene) or three-dimensionally (such as in diamond or sodium chloride). These substances are not composed of molecules. Solid metals are also not composed of molecules.



▲ **fig A** Dot-and-cross diagram for hydrogen with overlapping circles.

DISCRETE MOLECULES

A **discrete (simple) molecule** is an electrically neutral group of two or more atoms held together by covalent bonds.

DOT-AND-CROSS DIAGRAMS

Covalent and polar covalent bonding in discrete molecules can be shown by dot-and-cross diagrams.

Fig A shows the example of hydrogen, H_2.

Further examples of dot-and-cross diagrams are shown in **table A**.

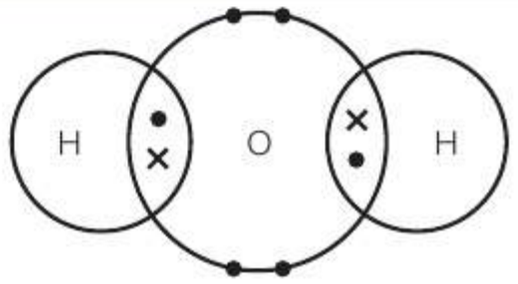
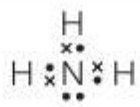
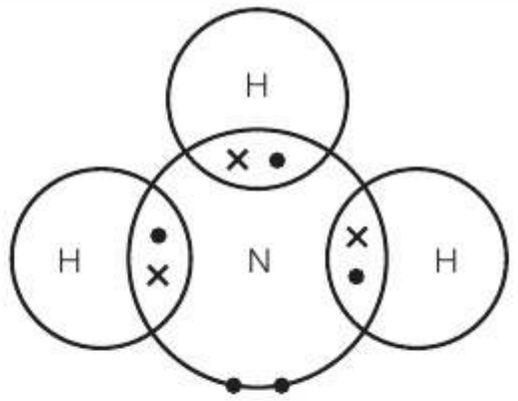
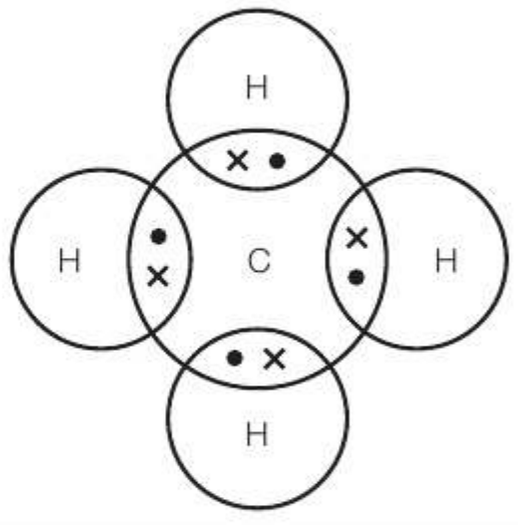
SUBSTANCE	DOT-AND-CROSS DIAGRAM	
Water, H_2O		
Ammonia, NH_3	 nucleus of hydrogen atom (electronegativity 2.1)	
Methane, CH_4		

table A Dot-and-cross diagrams for water, ammonia and methane.

EXAM HINT

When drawing a molecule such as chloromethane, do not forget to show all of the non-bonding electrons on the chlorine atom.

THE OCTET RULE

You might read that in order to form a stable compound, the outer shell of each atom must have the same number of electrons as the outer shell of a noble gas. In most cases this will be eight electrons. This has led to a rule that is often referred to as the 'octet rule'.

This is not always true, as you can see from the examples in **table B**. In each case, the outer shell of the central atom of the molecule does not contain eight electrons.

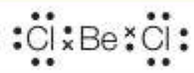
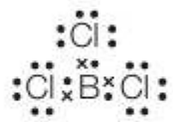
SUBSTANCE	DOT-AND-CROSS DIAGRAM	NUMBER OF ELECTRONS AROUND CENTRAL ATOM
Beryllium chloride, $BeCl_2$		4
Boron trichloride, BCl_3		6
Phosphorus(V) chloride, PCl_5		10
Sulfur hexafluoride, SF_6		12

table B Examples breaking the octet rule.

DOT-AND-CROSS DIAGRAMS OF MOLECULES CONTAINING MULTIPLE BONDS

Table C shows the dot-and-cross diagrams for three molecules (O_2, N_2, CO_2) that contain a double or triple bond.

DISPLAYED FORMULAE (FULL STRUCTURAL FORMULAE)

A **displayed (full structural) formula** shows each bonding pair as a line drawn between the two atoms involved.

Table C gives some examples of dot-and-cross diagrams together with the displayed formulae.

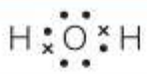
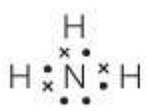
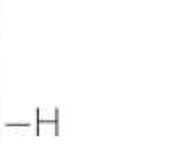
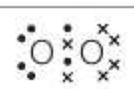
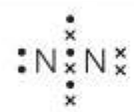
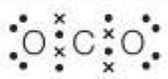
SUBSTANCE	DOT-AND-CROSS DIAGRAM	DISPLAYED FORMULA
Water, H_2O		H–O–H
Ammonia, NH_3		
Oxygen, O_2		O=O
Nitrogen, N_2		N≡N
Carbon dioxide, CO_2		O=C=O

table C Examples of displayed formulae with the corresponding dot-and-cross diagram.

LEARNING TIP

Although it is essential to show all of the non-bonding (lone) pairs of electrons in a dot-and-cross diagram, it is not necessary to show them in a displayed formula.

SUBJECT VOCABULARY

discrete (simple) molecule an electrically neutral group of two or more atoms held together by covalent bonds

displayed (full structural) formula a formula that shows each bonding pair as a line drawn between the two atoms involved

CHECKPOINT

1. Draw a dot-and-cross diagram for each of the following molecules:

(a) H_2S (b) PH_3 (c) PF_3 (d) SCl_2 (e) AsF_5 (f) HCN (g) SO_2

2. Draw the displayed formula for each of the molecules in Question 1.

3B 4 DATIVE COVALENT BONDS

SPECIFICATION REFERENCE 3.11(ii)

LEARNING OBJECTIVES

- Know that in a dative covalent bond both electrons in the bond are supplied by only one of the atoms involved in forming the bond.
- Be able to draw dot-and-cross diagrams for some molecules and ions that contain dative covalent bonds, including Al_2Cl_6 and the ammonium ion.

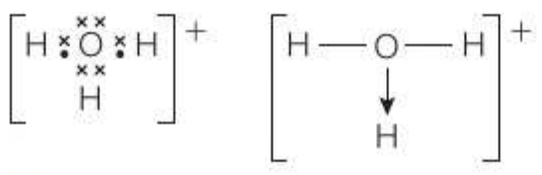
DATIVE COVALENT BOND FORMATION

A **dative covalent bond** is formed when an empty orbital of one atom overlaps with an orbital containing a non-bonding pair (lone pair) of electrons of another atom.

The bond is often represented by an arrow from the atom providing the pair of electrons, to the atom with the empty orbital. Below are three examples of dative covalent bonds.

THE HYDROXONIUM ION, H_3O^+

The dot-and-cross diagram and the displayed formula of a hydroxonium ion are shown in **fig A**.

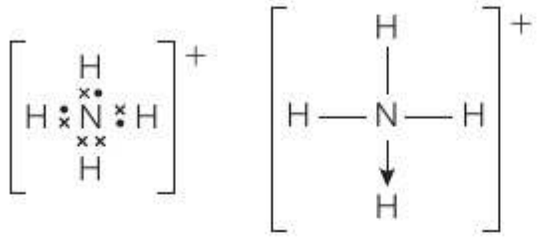


▲ **fig A** Dot-and-cross diagram and displayed formula for the hydroxonium ion.

The empty 1s orbital of the H^+ ion overlaps with the orbital of the oxygen atom that contains the lone pair of electrons.

THE AMMONIUM ION, NH_4^+

The dot-and-cross diagram and the displayed formula of an ammonium ion are shown in **fig B**.

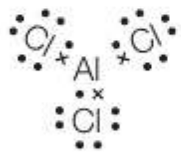


▲ **fig B** Dot-and-cross diagram and displayed formula for the ammonium ion.

The empty 1s orbital of the H^+ ion overlaps with the orbital of the nitrogen atom that contains the lone pair of electrons.

ALUMINIUM CHLORIDE, Al_2Cl_6

The aluminium atom in the $AlCl_3$ molecule has only six electrons in its outer shell and so has an empty orbital (**fig C**).

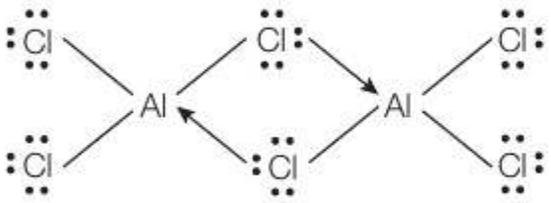


▲ **fig C** Dot-and-cross diagram for aluminium chloride.

In the gas phase, just above its sublimation temperature, aluminium chloride exists as Al_2Cl_6 molecules (**fig D**).

Two $AlCl_3$ molecules bond together. One of the atomic orbitals of a chlorine atom of one $AlCl_3$ molecule that contains a lone pair overlaps with the empty orbital of the aluminium atom of a second $AlCl_3$ molecule. The same happens between the chlorine atom of the second molecule and the aluminium atom of the first molecule.

One chlorine atom from each molecule acts as a bridge connecting the two molecules with dative covalent bonds.



▲ **fig D** Displayed formula for the aluminium dimer.

CHECKPOINT

1. (a) Draw a dot-and-cross diagram for a molecule of NH_3 and a molecule of BF_3.

(b) Draw a dot-and-cross diagram for a molecule of $NH_3 \cdot BF_3$.

2. Draw a dot-and-cross diagram and displayed formula for the $AlCl_4^-$ ion and identify the dative covalent bond.

3. One way of describing the bonding in a molecule of carbon monoxide (CO) is to state that it contains two covalent bonds and one dative bond. Using this description, draw a dot-and-cross diagram and displayed formula for a molecule of carbon monoxide.

SUBJECT VOCABULARY

dative covalent bond the bond formed when an empty orbital of one atom overlaps with an orbital containing a lone pair of electrons of another atom